PRACTICAL NO. 6

Interfacing a 16x2 LCD (I2C) with Raspberry Pi

Aim

Interface a 16x2 LCD module with Raspberry Pi to display user-defined messages on the screen.

Requirements

- Raspberry Pi 400
- LCD 1602 with I2C module (PCF8574 backpack)
- 4 female-to-female jumper wires

Circuit Connections (I2C LCD)

LCD Pin	Connect to Raspberry Pi GPIO
GND	Pin 6 (GND)
VCC	Pin 4 (5V)
SDA	Pin 3 (GPIO 2 - SDA)
SCL	Pin 5 (GPIO 3 - SCL)

Connection diagram:

```
LCD       Pi 400 GPIO
----      ------------
GND   --> GND (Pin 6)
VCC   --> 5V  (Pin 4)
SDA   --> SDA (GPIO 2 / Pin 3)
SCL   --> SCL (GPIO 3 / Pin 5)
```

Step 1: Enable I2C on the Raspberry Pi

```
sudo raspi-config
```

Go to **Interface Options > I2C > Enable**, then reboot the Pi.

Step 2: Install the required packages

```
sudo apt update
sudo apt install i2c-tools python3-smbus python3-pip
pip3 install RPLCD
```

On Raspberry Pi OS Bookworm, pip may require the extra option **--break-system-packages** (or use a virtual environment).

Check that the LCD is detected:

```
i2cdetect -y 1
```

Sample output:

```
     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:          -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- 27 -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
```

Note down the address shown (usually **0x27** or **0x3F**); here it is 0x27.

Step 3: Test the LCD with Python

Save the following program as **lcd_test.py**:

Code: lcd_test.py

```
from RPLCD.i2c import CharLCD
from time import sleep

# Change 0x27 if i2cdetect shows a different address
lcd = CharLCD('PCF8574', 0x27, cols=16, rows=2)

lcd.write_string("Hello from Pi 400!")
sleep(3)
lcd.clear()
```

Run it:

```
python3 lcd_test.py
```

Optional - display a user-defined message typed at the keyboard:

Code: custom_message.py

```
from RPLCD.i2c import CharLCD
from time import sleep

lcd = CharLCD('PCF8574', 0x27, cols=16, rows=2)

line1 = input("Enter line 1 (max 16 characters): ")
line2 = input("Enter line 2 (max 16 characters): ")

lcd.clear()
lcd.write_string(line1[:16])
lcd.crlf()
lcd.write_string(line2[:16])
sleep(5)
lcd.clear()
```

Output

The LCD displays the message as shown below.



Message displayed on the 16x2 I2C LCD